# Empowering Diabetes Care Through an Intelligent Knowledge Assistant

DISSERTATION

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1. Advanced Deep Learning
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**CERTIFICATE**

This is to certify that the Dissertation entitled **Intelligent Knowledge Assistant for Diabetes (IKAD)** and submitted by Ms.**Tanya** ID No. **2022ac05583**

in partial fulfillment of the requirements of DSECLZG628T / AIMLCZG628T Dissertation, embodies the work done by him/her under my supervision.

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## Abstract

Managing diabetes is a continuous and complex process that requires careful blood glucose monitoring, structured meal planning, strict medication adherence and regular physical activity. These tasks are not only time-intensive but also demand real-time, individualized guidance—for example, understanding how different foods affect glucose levels or adjusting medication after exercise. Despite the availability of digital diabetes management tools, most provide generalized recommendations that do not account for real-world variations such as meal timing, stress, or unplanned physical activity. As a result, many individuals struggle to follow treatment plans, make timely decisions and maintain effective glucose control, often leading to preventable health complications.

To address these challenges, this dissertation presents the Intelligent Knowledge Assistant for Diabetics (IKAD), an AI-driven system designed to provide real-time, evidence-based responses to diabetes-related queries.

The primary objective of IKAD is to reduce the burden of daily diabetes management by delivering practical, context-aware recommendations that help users make informed decisions—whether it involves choosing appropriate foods, modifying exercise routines, or adjusting medication dosages. Unlike existing solutions that rely on static rule-based approaches, IKAD dynamically tailors its suggestions based on both clinical guidelines and real-world contextual factors. By transforming broad recommendations into specific, personalized advice, IKAD empowers patients with customized care that directly aligns medical knowledge with everyday support.

Diabetes is a lifelong condition that demands continuous management, including regular blood sugar monitoring, meal planning, medication adherence and maintaining an active lifestyle. Although digital tools for diabetes management are increasingly available, most offer generic advice that does not adequately address the real-life complexities of living with diabetes, such as blood sugar fluctuations caused by meal timing, stress, or physical activity. This dissertation introduces IKAD as a solution that reduces daily management challenges by delivering personalized guidance to tackle practical and situational issues directly.

At its core, IKAD incorporates artificial intelligence techniques to ensure accuracy, reliability and relevance. Large Language Models (LLMs) enable natural language comprehension and coherent conversational interactions, making the system accessible for users with varying levels of medical knowledge.

To ensure precision and up-to-date medical accuracy, retrieval-augmented generation (RAG) is employed, connecting IKAD to trusted sources such as India's National Center for Disease Control, the American Diabetes Association, the CDC and peer-reviewed clinical research. For example, when a user asks, "What should I eat if my blood sugar is high after lunch?", IKAD retrieves the latest dietary guidelines for managing post-meal glucose spikes and offers advice on low-glycemic foods or appropriate insulin adjustments in line with established clinical standards.

Additionally, IKAD is designed to incorporate extra data sources, such as the user's medical history, glucose trends, dietary patterns and physical activity levels, to provide highly personalized recommendations. By considering factors like recent exercise or carbohydrate intake, the system refines its advice to ensure greater personal relevance and effectiveness.

Recognizing that diabetes is a lifelong condition, IKAD adapts its recommendations to support both immediate decisions and long-term treatment plans. To support scalability and long-term reliability, IKAD utilizes efficient indexing and retrieval mechanisms over large datasets, ensuring that it remains current with emerging research findings and evolving diabetes treatment protocols while handling high query volumes with consistent responsiveness.

By integrating AI methodologies, data analysis and personalized insights, IKAD bridges the gap between generalized medical advice and the specific, everyday guidance required for effective diabetes self-management. This system empowers patients to make informed, data-driven health decisions and provides accurate, personalized and actionable recommendations that contribute to improved long-term health outcomes.

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## Technological Foundations of IKAD

The Intelligent Knowledge Assistant for Diabetics (IKAD) is a tool engineered to provide real-time, context-sensitive, and evidence-based guidance for diabetes management.

Traditional digital tools often fall short by offering static, rule-based responses that fail to address the dynamic nature of diabetes care. IKAD overcomes these limitations by integrating cutting-edge artificial intelligence (AI), machine learning (ML) and data retrieval to deliver tailored recommendations. The system updates its responses using the latest clinical guidelines and research findings, ensuring that each recommendation is both accurate and immediately actionable.

IKAD's architecture is built on three main pillars:

1. Large Language Models (LLMs)
2. Retrieval-Augmented Generation (RAG)
3. Distributed Machine Learning and Scalablity

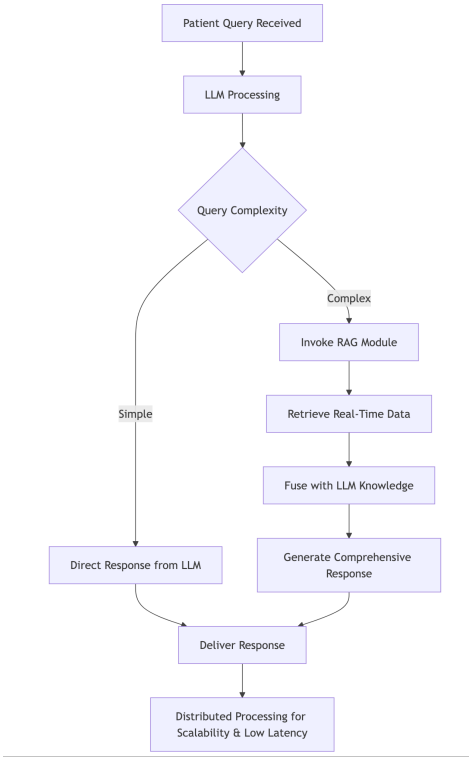


*Figure 1: Decision Flow for Patient Queries Based on Complexity*

1. **Large Language Models (LLMs): Natural Language Understanding and Response Generation**

IKAD's interactive capabilities are powered by a state-of-the-art Large Language Model (LLM), which has been specifically trained on extensive medical literature and clinical data. This model forms the foundation for understanding user queries and generating precise responses.

**(i) Deep Semantic Analysis and Natural Language Parsing:**

a. **Interpreting Complex Medical Queries:**
The LLM is designed to go beyond simple keyword detection. It performs deep semantic analysis to understand the underlying intent behind questions. For instance, it can distinguish between an urgent request ("Should I eat after taking insulin?") and an inquiry that seeks explanation ("How does insulin timing affect my meals?").

b. **Specialized Medical Training:**
By being fine-tuned on medical datasets, the LLM accurately recognizes technical terms and conditions, ensuring that even queries with subtle nuances are interpreted correctly. This results in responses that are both clinically accurate and context-specific.

**(ii) Contextual Memory and Multi-Turn Dialogue:**

(a) **Retention of Conversation History:** The model is capable of maintaining a context over multiple exchanges. When a user asks a follow-up question, such as "Would exercise help if my blood sugar is high?" after an initial query about high blood sugar, the system uses prior context to produce an answer that is coherent and connected.

(b) **Adaptive Dialogue:**
This memory function allows IKAD to adjust the level of detail and depth in responses as the conversation evolves, ensuring that every piece of advice remains consistent and responsive to the user's ongoing concerns.

**(iii)Transformer-Based Attention Mechanisms**

(a) **Focused Information Extraction:** Leveraging Transformer architecture, the LLM employs attention mechanisms to identify and focus on critical medical terms within a query. This results in responses that are both concise and information-rich.

(b) **Dynamic Complexity Adjustment:**The model automatically tailors its response based on the complexity of the query. It provides detailed explanations when needed or concise answers for simpler questions… all without losing technical accuracy.

**3. Retrieval-Augmented Generation (RAG): Real-Time Integration of Authoritative Medical Data**

While the LLM provides strong language comprehension, its inherent limitation is that its pre-trained knowledge may become outdated. IKAD's Retrieval-Augmented Generation (RAG) component addresses this by dynamically integrating current, authoritative data into the response process.

a. **Real-Time Data Retrieval from Trusted Sources:** RAG enables IKAD to fetch the latest medical guidelines and research findings from highly trusted sources such as the American Diabetes Association (ADA) and Centers for Disease Control and Prevention (CDC).

b. **Fusion of Pre-Trained and Real-Time Knowledge:** The RAG component fuses the pre-trained knowledge of the LLM with the dynamic data retrieved from external sources. This results in responses that are both grounded in established clinical practice and informed by the latest medical research.

4. **Scalability, Reliability and Rapid Response**

IKAD is designed to efficiently handle large volumes of queries while ensuring minimal resource overhead and scalability. Instead of relying on a single system, IKAD efficiently retrieves embeddings from FAISS cloud/Qdrant or cached results, while inference runs only when needed using a serverless API. This ensures that responses remain fast and efficient, without unnecessary compute overhead. We scale automatically by leveraging auto-scaling serverless deployments, removing the need for manual intervention to ensure consistency without excessive overhead.

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**Properties:**

1. **Personalization Through Contextual Data:** A defining feature of IKAD is its ability to deliver highly personalized recommendations by incorporating a wide range of user-specific data:
    (a) **History:** IKAD factors in an individual's chronic conditions, medication regimen to tailor guidance on dosage adjustments or dietary modifications.

    (b) **Lifestyle and Activity Patterns:** Inputs on recent physical activities, meal choices and stress levels help IKAD refine its advice for better practicality and immediacy.

    (c) **Actionable Insights:** By integrating diverse data sources, the system can pinpoint solutions that align with each patient's unique health circumstances, thereby increasing the likelihood of adherence and positive clinical outcomes.

2. **Scalability and Accessibility:** The platform's efficient indexing and retrieval methods ensure that users receive timely, accurate advice. By focusing on low-latency interactions, IKAD maintains responsiveness, a critical factor for continuous, day-to-day diabetes management.

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**Research Foundations and Innovations Underpinning IKAD**

The development and implementation of the Intelligent Knowledge Assistant for Diabetics (IKAD) draw on three primary research domains. Each domain contributes critical insights that influence IKAD's architecture, design choices, and operational efficacy:

1. Conversational AI in Healthcare
2. Personalized Tools for Chronic Disease Management
3. AI Techniques for Delivering Accurate Real-Time Responses

The following sections explore each domain in detail, examining the techniques that inform IKAD's functionality.

**1. Conversational AI in Healthcare**

Conversational AI in healthcare has been instrumental in transforming patient engagement and enhancing care delivery. Key aspects include:

**a. Building Patient Trust**
Patient trust hinges on the reliability and clinical accuracy of AI-generated medical advice. For conversational AI systems like IKAD to be effective, they must deliver precise, safe and trustworthy responses comparable to those of healthcare professionals and provides a foundational understanding of the essential attributes that IKAD must have to gain patient trust. IKAD's responses encourage users to follow the guidance provided, knowing that it is anchored in trusted medical practice.

b. **Ensuring Reliability of Responses**: Conversational systems in healthcare must avoid misinformation that can lead to unsafe medical decisions.
By integrating large language models (LLMs) with Retrieval-Augmented Generation (RAG), IKAD cross-references each generated answer with authoritative data sources. This dual-layer approach minimizes errors and ensures that every response is safe, clinically sound, and tailored to the patient's context.

**2. Personalization for Chronic Disease Management**

Effective management of chronic conditions like diabetes requires personalized interventions. The research in this domain highlights several key aspects:

a. **Lifestyle-Focused Interventions:** Wu et al. (2023) have shown that personalized healthcare tools, which adapt to an individual's lifestyle, yield significantly improved outcomes. IKAD gathers detailed user-specific data such as meal history, physical activity, stress levels, and daily routines to deliver recommendations that reflect the patient's unique lifestyle. This customization helps in fine-tuning interventions to prevent complications and promote long-term health.

b. **Predictive Capabilities for Proactive Management:** Zhu et al. (2021) emphasize the importance of predictive analytics in chronic disease management, enabling early intervention before severe episodes occur.
By integrating continuous monitoring data (like blood glucose trends) with historical health records, IKAD can forecast potential hypoglycemic or hyperglycemic events. This predictive functionality allows for timely preventive measures, transforming reactive care into proactive management.

3. **AI Techniques for Delivering Accurate Real-Time Responses:**

To maintain high-quality, real-time interactions, IKAD leverages several techniques. These techniques are categorized into three subdomains:

a. **Understanding Patient Queries: Contextual Parsing and Nuanced Interpretation:** The breakthrough by Vaswani et al. (2017) in Transformer-based models has revolutionized how systems understand context in natural language. IKAD's LLM uses attention mechanisms to capture the contextual relationships among words.
This enables the system to parse intricate queries- such as those involving precise medication timings or combined exercise recommendations with insulin adjustments, with a level of granularity that surpasses traditional keyword-based methods.

b. **Delivering Accurate and Trusted Responses (Integration of External Evidence via RAG):** The work of Liu et al. (2021) and Xiong et al. (2024) underscores the value of merging generative language models with real-time data retrieval.
IKAD's RAG module dynamically retrieves the information from authoritative sources and fuses it with the LLM's output. This dual mechanism guarantees that each response is evidence-based and that recommendations are continuously updated to reflect medical findings.

c. **Scalable Deployment with Distributed Machine Learning:** Froelicher et al. (2020) recommend distributed ML pipelines for scalable deployment in high-demand environments. IKAD's architecture distributes tasks across multiple computational nodes.

## List of Symbols & Abbreviations Used

| Abbreviation | Full Form |
|---|---|
| **ADA** | American Diabetes Association |
| **CGMs** | Continuous Glucose Monitors |
| **CDC** | Center for Disease Control |
| **FAISS** | Facebook AI Similarity Search |
| **GPR** | Gaussian Process Regression |
| **IKAD** | Intelligent Knowledge Assistant for Diabeties |
| **LLMs** | Large Language Models |
| **ML** | Machine Learning |
| **NLP** | Natural Language Processing |
| **Q&A** | Questions and Answers |
| **FAQ** | Frequently Asked Questions |
| **RAG** | Retrieval-Augmented Generation |
| **T5** | Text-to-Text Transfer Transformer |
| **D3.js** | Data-Driven Documents (JavaScript Library) |

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# Chapter 1: The Purpose of the Work and Expected Outcomes

## 1.1 Purpose of the Work and Overall Motivation

This research addresses the pressing need for precise, context-aware guidance in diabetes management. Although numerous applications collect user data and present generic information on glucose monitoring or dietary logs, such platforms often fail to deliver immediate, clinically aligned advice that users can act upon.

The Intelligent Knowledge Assistant for Diabetics (IKAD) narrows this gap by uniting advanced Natural Language Processing (NLP), specialized Deep Learning models, a distributed computing framework and a scalable deployment architecture. The system analyzes free-form user queries, consults authoritative guidelines, and produces evidence-based recommendations in real time. Through this design, IKAD transforms isolated diabetes-related inputs into practical clinical insights.

### 1.1.1 Contextual Challenges in Diabetes Management

Daily management of diabetes requires balancing medication regimens, dietary composition, and lifestyle factors such as exercise frequency. Small deviations in routine may cause significant shifts in blood sugar levels, leading to acute events that demand immediate intervention. Many existing digital tools record these fluctuations but rarely provide actionable recommendations that incorporate best-practice guidelines. This situation prolongs user dependence on sporadic clinical visits or generalized advice from static logs.

IKAD's framework seeks to address such shortcomings by integrating advanced AI to interpret nuanced questions, retrieving trusted medical references, and presenting verified, patient-specific recommendations.

### 1.1.2 Anticipated Contributions and Outcomes

The system delivers tailored answers to queries about insulin dosing, meal adjustments, and exercise planning while simultaneously ensuring that each recommendation aligns with recognized clinical guidelines (documents from the Centers for Disease Control (CDC) or the American Diabetes Association (ADA)). By harnessing powerful NLP strategies, deep learning architectures, and a robust distributed infrastructure, IKAD provides a platform that can scale with user demand without sacrificing accuracy or speed. As a result, it holds potential for improving self-management outcomes in individuals with diabetes, reducing episodes of severe blood sugar imbalance, and easing clinical burdens through timely, AI-driven support.

## 1.2 Pillars of the Approach

### 1.2.1 Advanced Natural Language Processing and Retrieval-Augmented Generation

IKAD's foundational pillar lies in its capacity to parse and respond to complex natural language queries about diabetes care. A multi-turn conversational system preserves context across successive user interactions, enabling more coherent responses. The system's core NLP engine (based on Transformer architectures) benefits from a process called Retrieval-Augmented Generation (RAG). When a user poses a question, the language model first constructs a preliminary answer. The RAG module then retrieves pertinent passages from indexed diabetes guidelines or reference documents, cross-checking the model's draft. Any discrepancies are identified and reconciled, ensuring that final outputs conform to recognized medical standards.

This synergy between generative and retrieval components significantly curtails risks of "hallucination" or outdated recommendations.

### 1.2.2 Specialized Deep Learning Model Design

The second pillar emphasizes sophisticated deep learning mechanisms that move beyond surface-level question-answering. While the foundational Transformer models are initially pre-trained on vast textual corpora, IKAD refines their clinical precision through parameter-efficient methods such as Low-Rank Adaptation (LoRA). By applying a minimal learning rate and restricting training epochs, these techniques delicately incorporate domain-specific reasoning into the model's parameters.

The process ensures that the system's broad linguistic competence remains intact, augmented by improved accuracy on clinically oriented tasks. This refined model handles nuanced queries related to insulin timing, carbohydrate ratios, and potential interactions with other medications or dietary choices.

### 1.2.3 Distributed Computing for Performance and Reliability

High user concurrency and the need for real-time responses demand an infrastructure that can handle large computational loads without sacrificing speed. IKAD's distributed framework parallelizes inference tasks across multiple compute nodes or GPUs. This design enables the system to process numerous simultaneous queries, maintaining responsiveness even under surges in usage. Containerized microservices split key tasks- such as language model inference, guideline retrieval, and user-facing interfaces- into distinct services. Each service runs independently but coordinates through well-defined interfaces. Should one service become overloaded, orchestrators can scale it out horizontally by launching additional replicas. This ensures low-latency interactions, which is particularly important in a clinical decision-support context where delayed recommendations might lead to dangerous health outcomes.

### 1.2.4 Scalable Deployment Architecture

Complementing the distributed computing backbone is an architectural commitment to seamless scalability and maintainability. IKAD's microservices, wrapped in lightweight containers, can be deployed on cloud-native platforms that automatically manage resource allocation, failover recovery, and rolling updates. Standard orchestration tools simplify the process of monitoring system health and adjusting compute resources in real time, eliminating single points of failure. This strategy benefits both research-oriented and clinical use cases: large healthcare networks with heavy user traffic, as well as smaller practices needing a streamlined solution, can both adopt IKAD. Continuous integration and deployment pipelines automate testing and rollout of new model versions, ensuring that improvements—such as updates to diabetes guidelines—propagate swiftly without interrupting the user experience.

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## 1.3 Important Factors in Diabetes Care

### 1.3.1 Medication Titration and Glucose Dynamics

Medication management in diabetes requires precise calibration. Factors such as meal composition, stress, and physical activity can substantially alter a patient's insulin or oral hypoglycemic requirements. By interpreting user questions about dosage changes or timing adjustments, IKAD's deep learning engine can offer informed suggestions verified against established medical references. This mechanism can help reduce episodes of hypoglycemia or hyperglycemia by aligning treatment modifications with patterns recognized in authoritative diabetes guidelines.

### 1.3.2 Nutritional Strategies and Lifestyle Modifications

Nutritional adjustments often pose a challenge. Patients vary widely in their cultural food preferences, daily schedules, and metabolic responses. IKAD consolidates these variables by leveraging advanced question-answering modules that reference accepted dietary standards and adapt them to individual circumstances. Coupled with RAG-driven verification, the system ensures that any guidance (eg. reducing carbohydrate portions in certain meals or including specific fiber sources) remains consistent with existing clinical evidence. Moreover, individuals seeking to integrate physical exercise can consult IKAD about pre-exercise carbohydrate intake or post-exercise monitoring intervals, thus obtaining scenario-specific advice grounded in deep learning inferences.

### 1.3.3 Continuous Risk Assessment

Although IKAD does not rely on streaming sensor feeds, it accommodates the concept of risk thresholds by examining previously reported glucose values and typical patterns of fluctuation. Whenever the user queries about potential concerns- like unusual fatigue, unexpected glucose spikes, or side effects- the NLP model interprets the question's nuances, and the retrieval module brings forth guideline excerpts relevant to acute interventions or escalation pathways. This design aims to mitigate overlooked risks and provides immediate cautionary steps, ensuring that any advice aligns with documented best practices in diabetes management.

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### 1.4 Common Gaps in Traditional Health Applications

The table below contrasts widespread limitations in contemporary diabetes apps with IKAD's intended solutions:

| Observed Shortcoming | Impact on Diabetes Care | IKAD's Resolution |
|---|---|---|
| Reliance on Generic or Pre-Programmed Tips | Users receive one-size-fits-all advice that may not match their individual conditions | Customized deep learning inferences, refined by domain-specific QA data, yield recommendations that adapt to each user's medication routine and dietary profile |
| Absence of Real-Time Guidance Verification | Potential for outdated or speculative answers | Retrieval-Augmented Generation cross-checks the system's outputs with guidelines, ensuring alignment with validated medical recommendations |
| Scalability Issues in High-Usage Scenarios | Lengthy response times or service outages | Distributed architecture coupled with containerization preserves responsiveness and resilience even under traffic spikes |
| Fragmented Upgrades to AI Models | Delayed availability of improved or corrected advice | Seamless CI/CD pipelines apply incremental model enhancements without forcing downtime or relying on specialized DevOps expertise |

By addressing these persistent gaps, IKAD stands poised to redefine what "digital support" can look like for individuals managing diabetes daily.

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### 1.5 System Architecture and Workflow Explanation

In practical use, a query travels through several core steps before the user receives a validated recommendation. This architecture ensures that every query benefits from both the generative strengths of a large-scale language model and the factual rigor of well-documented medical standards.

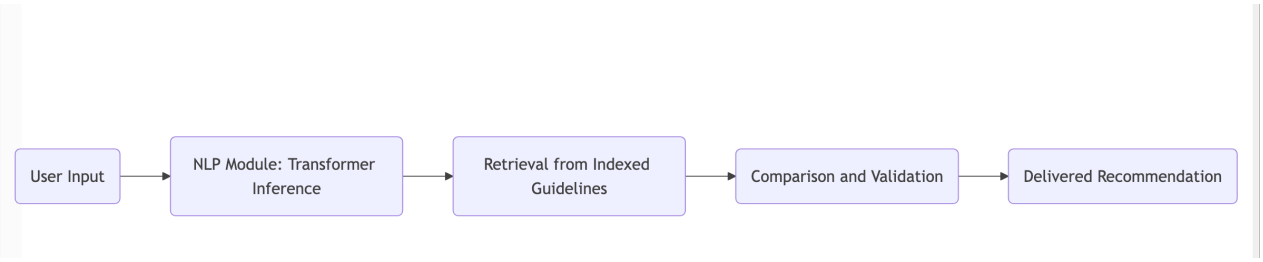


*Figure 2: High-Level Flow of NLP Inference and Guideline Retrieval*

1. A user poses a question, often containing specific context—such as a recent medication change or an intention to start a new exercise routine.
2. IKAD's NLP engine, powered by a Transformer-based deep learning model, interprets the query's details and generates an initial response.
3. Concurrently, the retrieval module scans its indexed repository of CDC/ADA guidelines, extracting text passages relevant to the question's scope.
4. A comparison stage flags potential mismatches between the model's draft and the retrieved references. Any discrepancies are reconciled so that the finalized guidance is consistent with reliable, authoritative sources.

5. The verified recommendation, complete with relevant detail or cautionary advice, is returned to the user in real time.

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### 1.6 Evaluation and Fine-Tuning Methodology

#### 1.6.1 Dataset Composition and Relevance

To ensure that IKAD's deep learning modules operate effectively in varied clinical scenarios, a specialized dataset is employed during fine-tuning to maintain semantic accuracy.

#### 1.6.2 Fine-Tuning Process (Low-Rank Adaptation)

The fine-tuning strategy balances the infusion of specialized knowledge with the preservation of overall language fluency. IKAD accomplishes this using LoRA, which updates only a small proportion of the language model's parameters at a low learning rate. Limiting the number of training epochs further prevents overfitting and preserves the model's general-purpose conversation skills. In effect, this approach adds nuanced diabetes reasoning to the system's inference process without diminishing its ability to understand the wide range of phrasing typical of user queries.

#### 1.6.3 Verifying Practical Efficacy

Beyond automated scores, targeted user tests examine how effectively IKAD handles real-world scenarios. These tests involve hypothetical user sessions dealing with fluctuations in blood glucose, medication adjustments, or daily meal variations. A correct answer requires not only that the system produce text aligned with best practices, but that it also present the information in a form that addresses the user's immediate concerns. The synergy between advanced NLP, RAG verification, and a well-structured dataset forms the backbone of this qualitative evaluation, ensuring that recommendations are both accurate and actionable in practical contexts.

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### 1.7 Operational Deployment and Scalability

From the perspective of computational deployment, IKAD adopts container-based microservices. Each container includes only the resources required for its respective function—NLP processing, retrieval, orchestration, or user interfacing. Orchestrators monitor key performance indicators (such as memory usage, average response time, and container health) and can launch additional service replicas to handle spikes in incoming requests. Because these containers communicate via standardized protocols, scaling out can occur independently for the Transformer inference module or the retrieval service, depending on which subsystem experiences the heavier load.

A continuous integration and deployment pipeline ensures that refined model weights, updated guidelines, or bug fixes can be introduced seamlessly. This mechanism avoids manual and often error-prone updates to a central system, which might disrupt continuity of care. Instead, it unifies version control of code, model checkpoints, and container configurations, lowering operational risk in dynamic research or clinical environments.

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### 1.8 Illustrative Scenarios of System Usage

#### 1.8.1 Insulin Adjustment Prompt

Consider a user experiencing frequent late-afternoon hypoglycemic readings. They submit a query describing their history of glucose readings and dosing schedule. The NLP module scrutinizes the question's details, identifies relevant guidelines related to insulin titration, and compares the generative output with guideline excerpts from the retrieval index. A final recommendation might suggest incrementally lowering midday insulin dosage or advising a mid-afternoon snack, carefully referencing the threshold patterns validated in official recommendations.

#### 1.8.2 Exercise and Diet Coordination

Another user might inquire about balancing a new exercise routine with their existing medication regimen. IKAD's approach analyzes the type of activity (say aerobic exercise) and the user's typical glycemic responses. The system then retrieves guidelines on pre-workout carbohydrate intake to prevent hypoglycemic dips. It merges that information with the user's historical patterns as gleaned from prior queries, producing targeted adjustments like slightly lower pre-exercise insulin or a recommended carbohydrate snack before activity begins.

#### 1.8.3 Adherence and Lifestyle Queries

Some queries revolve around broader lifestyle issues, such as shifting meal times or dealing with stress-induced glucose variability. IKAD's deep learning model interprets these nuanced questions, forms a cohesive answer, and cross-references authoritative psychological or lifestyle recommendations that appear in the retrieved guidelines. Users receive a consolidated overview that ties daily routines, stress management techniques, and dietary changes into a coherent plan.

| Situation | Patient Query | IKAD's Response |
|---|---|---|
| **Unexpected Hypoglycemia** | "I feel shaky, and my blood sugar is 60 mg/dL. What should I do?" | "Consume 15 grams of fast-acting carbs, like fruit juice. Recheck your glucose after 15 minutes and repeat if low." |
| **Bedtime Blood Sugar Concerns** | "My blood sugar is 120 mg/dL before bed. Should I eat something?" | "Your levels are safe, but to avoid overnight drops, consider a light snack like peanut butter on whole-grain bread." |
| **Post-Exercise Recovery** | "I walked more today, and now my glucose is 70 mg/dL. Should I eat?" | "Yes, eat 15 grams of fast-acting carbs, like a small apple, to stabilize your levels after unplanned activity." |
| **Stress Management** | "Work stress has pushed my blood sugar over 200 mg/dL. What can I do?" | "Take a 10-minute break for deep breathing or light walking. Stick to low-carb meals until levels stabilize." |

### 1.9 Future Directions and Potential Enhancements

An immediate area for extension lies in broadening the range of guideline documents and medical research indices that IKAD can parse, thereby expanding the set of verifiable references beyond CDC and ADA sources. Over time, the system might incorporate interpretability layers so that users and clinicians can see the rationale behind each step of the recommendation, offering a measure of transparency often lacking in black-box AI systems. Research could also investigate forms of domain adaptation- potentially even user-specific tuning- where the system progressively refines its advice as it interacts with each individual's evolving clinical profile.

Additionally, an optional feedback loop from user interactions, once ethically reviewed and anonymized, could feed into incremental re-training or re-ranking models, making IKAD's suggestions increasingly personalized over time.

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By combining NLP techniques, a carefully fine-tuned deep learning backbone, robust distributed computing, and a scalable deployment approach, IKAD represents a transformative solution for diabetes decision support. The system's multi-turn conversation feature guides users through complex queries, while the RAG pipeline enforces alignment with authoritative clinical knowledge. Throughout the research, a balanced fine-tuning methodology ensures that the model's domain-specific performance improves without sacrificing its inherent language fluency. Meanwhile, a containerized, orchestrated architecture promises responsiveness and reliability under diverse load conditions.

In this way, IKAD stands as a blueprint for next-generation AI systems in health applications, where real-time, evidence-based recommendations can significantly improve outcomes and reduce the cognitive burden of disease management. Whether deployed in large-scale clinical settings or smaller practices, its flexible design-coupled with a deep respect for medical guidelines-underscores the viability of a unified framework that integrates advanced AI methods with the practical realities of daily diabetes care.

## Chapter 2: Model Development and Enhancement:

The goal is to provide accurate, context-aware, and clinically valid recommendations and insights to individuals managing diabetes using the IKAD (Intelligent Knowledge Assistant for Diabetes). Our assistant is trained and fine-tuned on a specialized model called Diabetica- derived from Qwen2- to achieve domain-specific expertise in diabetes care.

Initially, we experimented with T5 (Text-to-Text Transfer Transformer) to address a variety of tasks such as question answering, short medical instructions, and summarizations in the diabetes context.

Although T5 offered a unified text-to-text framework, it failed to maintain extended dialogue context and lacked the necessary domain depth for complex diabetes management tasks.

Consequently, we transitioned to a more specialized architecture inspired by the Diabetica family of models. The Diabetica approach focuses on domain-specific data curation, fine-tuning strategies, and benchmarking against real-world diabetic patient interactions and specially crafted datasets to ensure clinical accuracy and contextual relevance.

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### 2.1 Introduction

In diabetes management, large language models (LLMs) face challenges such as maintaining context in multi-turn conversations, correctly interpreting unstructured queries, and providing actionable, safe, and domain-specific recommendations. Our efforts to create a user assistant (IKAD) underscore the importance of an LLM that can handle the complexities of real-world clinical scenarios.

The Diabetica paper by Lai Wei et al. addresses these same complexities, emphasizing that standard MCQ-based evaluations (MedQA, MedMCQA, PubMedQA etc.) do not fully capture the interactive, skill-based nature of clinical tasks.

Their work introduced MedQA-CS, which focuses on Step 2 Clinical Skills (CS) exam-type interactions: information gathering, physical exams, patient closure, and differential diagnosis. This broader view aligns with our IKAD objectives.

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## 2.2 Background and Initial Experiments

### 2.2.1 Overview of the T5 Model in a Clinical Context

The T5 model (Text-to-Text Transfer Transformer) provided a uniform framework for various NLP tasks, converting each task into a text generation problem. Early in our project, we fine-tuned T5 on a dataset derived from PubMed articles, clinical guidelines, and simulated patient–doctor conversations to adapt it for diabetes management.

**Observations with T5**:

- **Context Handling**: While T5 produced intelligible, coherent responses for simple, structured queries, it struggled with long, multi-turn dialogues.
- **Genericity**: The model often defaulted to general medical statements, lacking the domain-specific depth crucial for precise diabetes care.
- **Unstructured Queries**: Real diabetic inquiries frequently include incomplete details and implicit assumptions. T5's responses were inadequate in responding to open-ended or multifaceted patient questions.

Given these limitations, T5 was not ideal for the dynamic, context-rich scenarios in diabetes management. These challenges motivated a shift toward a specialized approach.

### 2.2.2 Shortcomings of the T5 Approach

Despite domain-specific fine-tuning, T5 exhibited:

1. **Contextual Fragmentation**
   The model failed to preserve continuity over extended dialogues, resulting in fragmented responses.
2. **Generic Language Output**
   The model relied heavily on generic phrasing, providing insufficient precision for clinical tasks requiring detail, like medication titration or symptom assessment.
3. **Handling Unstructured Queries**
   In real-world diabetes management, patient input can be unstructured and full of subtext. T5's general approach struggled to interpret these nuances, leading to incorrect or incomplete advisories.

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### 2.3 Transition to the Diabetica 1B Model (Qwen2 Architecture)

#### 2.3.1 Rationale

Given T5's limitations in context retention and domain specificity, we shifted to a custom diabetes-centric approach. Diabetica 1B, based on the Qwen2 architecture, addresses three key objectives:

1. **Domain-Specific Pre-Training**
   Pre-training on data reflecting the clinical nuances of diabetes management: guidelines, patient interactions, and specialized knowledge.

2. **Extended Context Retention**
   Qwen2 is designed to handle longer sequences and multi-turn dialogues, critical for dynamic patient interactions.

3. **Precise Clinical Information**
   Fine-tuning ensures accurate terminology, correct dosing instructions, and alignment with recognized treatment standards.

#### 2.3.2 Technical Details of the Qwen2 Architecture

Qwen2 extends the classic Transformer with modifications for longer contexts and more efficient inference:

- Grouped Query Attention (GQA): Improves KV cache usage and enhances throughput over large sequences.
- SwiGLU Activation and Rotary Positional Embeddings: Capture sequence information more effectively, improving model capacity.
- RMSNorm Pre-Normalization: Stabilizes training for deep transformer stacks.

#### 2.3.3 Fine-Tuning Diabetica

Following the approach in Lai Wei et al. (Diabetica paper), we decided to go forward with Diabetica-1.5B, a lighter variant for resource-constrained scenarios, maintaining the domain adaptations for the purpose of fine-tuning to meet our requirements. This was done using:

1. **LoRA Parameter-Efficient Fine-Tuning**
   This reduces computational overhead and effectively retains the original LLM's knowledge.

2. **Self-Distillation**
   Seed LLM refines its own output which addresses catastrophic forgetting by narrowing the gap between newly introduced domain knowledge and the model's existing general knowledge.

### 2.4 Motivation from the Diabetica Paper and Dataset Overview

Noting the limitations of standard multiple-choice benchmarks to assess the interactive and skill-based tasks in medicine, the authors introduce an evaluation framework called MedQA-CS (inspired by the USMLE Step 2 CS and OSCEs).

Diabetica framework goes beyond knowledge recall to test models in:

- **Information Gathering**
- **Physical Examinations**
- **Patient Closure**
- **Differential Diagnosis**

Their MedQA-CS dataset provides richer clinical interactions, aligning well with our aim of building a contextually aware diabetes advisor (**IKAD**).

#### 2.4.1 Dataset Composition

The Diabetica team curated an extensive diabetes-focused dataset, drawing upon:

- **Public Multiple-Choice Question Banks**
  Using MCQ datasets (MedQA, MedMCQA, MMLU, CMMLU, CMB, CMExam) derived from USMLE and other advanced medical exams to ensure coverage of fundamental and complex diabetes topics.

- **Public Medical SFT (Supervised Fine-Tuning) Datasets**
  Incorporating open-source sets such as CMtMedQA, ChatMed, cMedQA2, DISC-Med-SFT to align the LLM with human-instructed medical knowledge.

- **Endocrinology Guidelines and Textbooks**
  Including diabetes guidelines, standard textbooks, and the DiaKG knowledge graph from Chinese diabetes resources (41 guidelines + expert consensus).

- **Drug Labels**
  Sourcing from a Chinese drug label repository to provide extensive coverage of anti-diabetic medications: indications, dosages, contraindications, side effects etc.

- **Real-World Dialogues**
  About 100 diabetes specialty questions answered by endocrine specialists, capturing authentic clinical interactions and language patterns.

### 2.4.2 Data Processing

The raw data undergoes multiple steps:

1. **Keyword Filtering**
   A set of positive keywords (like diabetes, HbA1c, blood sugar) and negative keywords ensures we retain only relevant diabetes-focused content.

2. **Deduplication:**
   Using SemDeDup with embeddings (bge-large-zh-v1.5) and k-means clustering, the dataset is cleansed of duplicates. Similar data points above a cosine similarity threshold are merged, retaining the longer, more detailed entries.

3. **Data Augmentation:**

   - Long Texts (guidelines, textbooks, drug labels) are split into segments. GPT-4 generates dialogues and fill-in-the-blank data, expanding training diversity.

   - MCQs are merged into coherent Chinese questions and refined with ChatGPT-3.5. GPT-4 then creates chain-of-thought reasoning to ensure correctness.

4. **Self-Distillation for Data Refinement**
   To mitigate catastrophic forgetting, given that the domain data might diverge from the LLM's original training, the authors use self-distillation:

   - Seed LLM generates an initial response.
   - A refinement prompt asks the same LLM to improve its own output.
   - The refined response replaces the original in the training set.

The result is MedQA-CS with 1,667 instruction-response pairs, categorized into InfoGatherQA, Physical Exams, Closures, and Differential Diagnosis.

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### 2.5 Benchmarking with the MCQ Dataset from Diabetica:

- MCQ Accuracy Metrics: Type A1 (Basic) vs. Type A2 (Case Study): 88.09% vs. 84.42%. (Outperforms open-source peers on these domain-specific questions)

- Fill-in-the-Blank (FB) Tasks: BERTScore=0.9298

- Text generation closely matches reference answers, indicating high semantic fidelity.

- Dialogue Benchmark (Single-Round Dialogue): open-ended responses were scored by GPT-4 and Claude-3.5, yielding 7.81 and 7.96 out of 10- better than base Qwen2-7B's 7.50 and 7.74.

- Catastrophic Forgetting and General Benchmarks: In GSM8K (math reasoning), Diabetica retained 99.6% of baseline performance. While in MMLU (multi-domain knowledge), there was slight improvement from 67.08 to 68.62.

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## 2.8 IKAD: System Architecture, Data Composition and Fine-tuning

### 2.8.1 Dataset Composition and Curation

Our base dataset to fintune and refine Diabetica for our purpose was built from:

- Filtered Patient Records Datasets: Real-world, anonymized data capturing genuine queries filtered for diabetes.
- Simulated Dialogues: Synthetic multi-turn exchanges to bolster the model's conversation handling (synthetic clinical QA).

All content underwent standardization of medical terminology, removing duplicates (SemDeDup) and proven alignment with current diabetes management standards. Note that the Diabetica needs to be adapted and personalized to handle English queries and the Q/A flow of an IKAD, hence the finetuning was necessary.

To make it coherent and better suited to our use case, we improved upon the above to create 1,700 prompts, each reflecting realistic clinical scenarios:

- Explicit Clinical Context: Includes patient's blood sugar history, recent meals, current medications, etc.

- Instructional Format: Guidance for the model to propose coherent, stepwise solutions.

- Alignment: ensuring no deviation from recognized medical protocols.

### 2.8.2 Fine-Tuning Methodology

1. Initialization: Starting with a pre-trained Diabetica 1B (Qwen2 architecture).

2. Sequential Prompt Application: The 1,700 prompts were fed iteratively.

3. Iterative Adjustments: Any incorrect or incomplete responses triggered prompt refinement.

4. Benchmark Integration: MCQ checks at each stage to maintain consistent clinical accuracy.

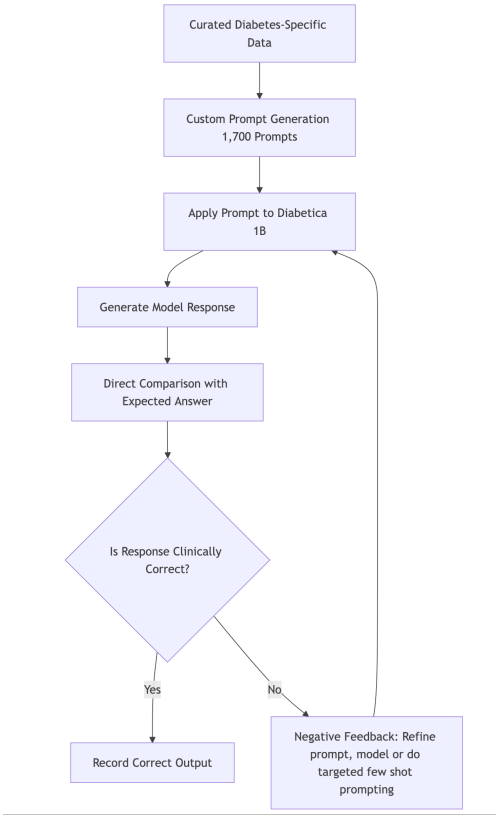


*Figure 3 Fine-Tuning Diabetica 1B with Curated Prompts*

The fine-tuning methodology was designed to maximize the model's ability to produce precise, contextually rich clinical responses. This process is summarized in the following steps:

1. **Data Pre-processing:**
   All input data was normalized to ensure consistency in terminology and format. Medical terms were standardized, and any ambiguous abbreviations were expanded into their full forms. This pre-processing stage was critical for aligning the data with the clinical language used in practice.

2. **Custom Prompt Generation:**
   The set of 1,700 prompts was developed in collaboration with clinical experts. Each prompt was designed to mimic a realistic clinical query, complete with detailed patient context. Examples include scenarios such as "A patient with elevated post-meal blood glucose seeks advice on dietary adjustments" or "A patient with a history of hypoglycemia in the morning asks for guidance on medication timing." These prompts provide explicit context without resorting to any extraneous or overly technical language, ensuring that the model's outputs remain strictly focused on clinical accuracy.

3. **Fine-Tuning Execution:**
   The Diabetica 1B model was fine-tuned using the curated dataset and the custom prompts. The training process involved iterative cycles where the model's outputs were compared against expert-provided answers. Any deviation from clinical guidelines prompted a revision of the corresponding prompt. This iterative process ensured that the final model was finely calibrated to produce responses that meet clinical standards.

4. **Validation Against MCQ Data:**
   At multiple stages during fine-tuning, the model was evaluated using the MCQ dataset from the Diabetica paper. This evaluation was done by directly inputting the MCQ questions and comparing the model's responses to the benchmark answers. Each evaluation cycle contributed to further prompt refinements, reinforcing the alignment between the model's output and the clinical objectives.

### 2.8.3 Benchmarking Workflow

The benchmarking analysis provides a rigorous quantitative and qualitative evaluation of the Diabetica 1B model's performance.

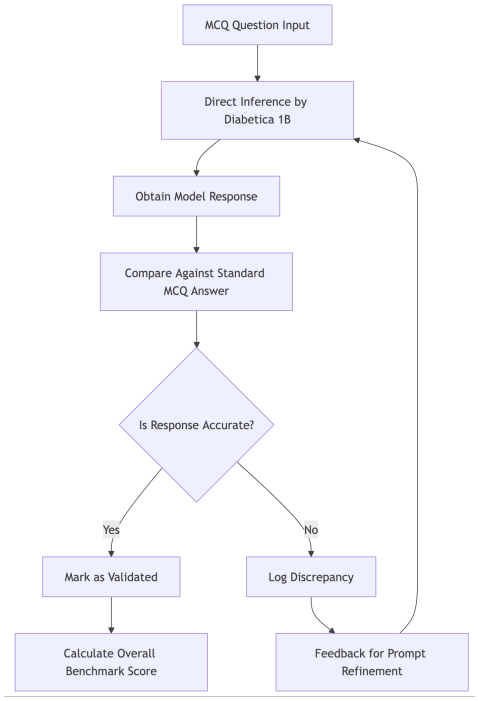


*Figure 4: MCQ Benchmarking Pipeline for Diabetica 1B*

Key aspects include:

- **Accuracy Assessment:**
  Each response generated by the model was scored based on its clinical correctness. The assessment criteria included the proper use of medical terminology, adherence to established guidelines, and the logical coherence of the response. The results were aggregated to produce an overall accuracy metric, which consistently showed that Diabetica 1B outperformed the baseline T5 model in terms of clinical validity.

- **Coherence and Context Retention:**
  Evaluators reviewed multi-turn dialogues to assess whether the model maintained context throughout the interaction. The Diabetica 1B model demonstrated a high level of coherence, effectively linking information across dialogue turns. This was critical for ensuring that the advice provided was not only accurate in isolated instances but also consistent over time.

- **User-Centric Relevance:**
  Although this evaluation was conducted in a controlled setting, the feedback from clinical experts confirmed that the model's responses were directly applicable to real-world scenarios. The relevance of the advice was measured by its ability to address the specific concerns presented in each query.

- **Direct Comparison with Benchmark Answers:**
  For each MCQ, the model's output was compared against the provided standard answer. Any discrepancies were meticulously documented and analyzed. The analysis revealed that the differences were largely due to minor linguistic variations rather than a fundamental misunderstanding of the clinical content.

- Moreover, a detailed analysis was conducted to compare the performance of Diabetica 1B with that of the T5 model. The following table summarizes the key metrics based on the MCQ benchmarking and qualitative evaluations:

| **Evaluation Metric** | **T5 Model** | **Diabetica 1B (Post Fine-Tuning)** |
|---|---|---|
| Clinical Accuracy (Qualitative) | Moderate, inconsistent | High, consistent with clinical guidelines |
| Multi-Turn Context Retention | Significant loss of context | Robust retention, high coherence |
| Relevance to Unstructured Queries | Variable, often too generic | Consistently relevant and detailed |

  Diabetica 1B demonstrates robust, context-driven outputs; T5 often lost continuity. Additionally, Diabetica 1B handles open-ended patient queries more accurately and Qwen2's architecture and domain focus enable longer, more coherent dialogues.

To illustrate the effectiveness of our approach, below are the analyses of specific clinical scenarios as encountered during the benchmarking process.

**Case Study 1: Dietary Adjustments for Post-Meal Hyperglycemia**

In this scenario, the prompt described a patient with elevated blood glucose levels following a meal. The prompt included details such as the time of the meal, the types of food consumed, and the patient's recent medication history. The expected clinical response was to advise on dietary modifications, including portion control and the inclusion of low-glycemic index foods.

- T5 Model Response:
  The T5 model provided a generic recommendation that did not adequately consider the patient's unique context. It mentioned general dietary guidelines but failed to offer specific advice tailored to the patient's condition.
- Diabetica 1B Response:
  The Diabetica 1B model, after fine-tuning, generated a detailed response that incorporated the specific timing and content of the meal. It recommended concrete dietary changes aligned with clinical guidelines, such as substituting high-glycemic foods with fiber-rich alternatives and scheduling follow-up assessments.

**Case Study 2: Managing Morning Hypoglycemia**

This scenario involved a patient experiencing hypoglycemia in the morning. The prompt provided context regarding the patient's overnight medication regimen and recent blood glucose measurements. The clinical objective was to adjust the patient's medication timing and dietary intake.

- T5 Model Response:
  The T5 model's output was generic and did not adequately address the need for adjusting the medication schedule. The response lacked specificity in terms of timing and actionable recommendations.
- Diabetica 1B Response:
  The Diabetica 1B model generated a response that explicitly mentioned the need to re-evaluate the overnight medication regimen. It provided step-by-step recommendations on how to adjust medication timing and included advice on an appropriate pre-bedtime snack to prevent nocturnal hypoglycemia.

**Case Study 3: Comprehensive Management in a Complex Scenario**

In a more complex scenario, the prompt described a patient with fluctuating blood glucose levels, inconsistent dietary habits, and variable physical activity. The challenge was to provide a coherent set of recommendations that addressed all these factors simultaneously.

- T5 Model Response:
  The T5 model's response was fragmented, addressing some aspects of the query while neglecting others. The overall recommendation was disjointed, leading to potential confusion for the patient.
- Diabetica 1B Response:
  The Diabetica 1B model provided a cohesive and integrated response. It broke down

the recommendations into clearly defined sections: dietary management, physical activity, and medication adherence. Each section included specific, actionable advice that was directly tied to the patient's reported data.

### 2.9 Error Analysis and Model Limitations

An error analysis was conducted to identify and quantify the remaining limitations of the Diabetica 1B model after fine-tuning. This section presents a detailed breakdown of error types, their frequency, and the contextual conditions under which they occur.

1. Contextual Loss Errors:
    - Occurrences where the model fails to maintain information from earlier dialogue turns, resulting in partial or fragmented responses.
    - These errors were predominantly observed in multi-turn interactions where the model was expected to reference previous patient data.

2. Specificity Errors:
    - Instances where the model generated responses that were overly generic or did not sufficiently tailor recommendations to the clinical scenario.
    - Commonly seen in queries with a high degree of variability, such as those involving complex patient histories or multiple co-morbidities.

3. Terminology Mismatch Errors:
    - Cases where the model misinterpreted or incorrectly applied specialized medical terminology.
    - Often observed when patient records included ambiguous abbreviations or region-specific terms not fully standardized in the training data.

**Case Studies:**

**(a) Extended Dietary Consultation**
In a scenario involving a detailed post-meal analysis, the model occasionally omitted references to earlier dietary inputs. On review, it was noted that errors correlated with instances where the patient's data spanned more than four dialogue turns. This observation suggests that while the Qwen2 architecture substantially improves context retention, there remains a diminishing return when the dialogue length exceeds a critical threshold.

**(b) Ambiguous Terminology in Medication Guidance**
In several instances, patient queries containing ambiguous medication abbreviations resulted in responses that either misinterpreted the abbreviation or used an incorrect clinical term. Although the curated dataset was pre-processed for standardization, residual inconsistencies in the original source data contributed to these errors.

(c) **Multi-Factor Clinical Recommendations**
Complex cases requiring integration of dietary, physical activity, and medication history occasionally produced segmented responses. In these cases, the model generated a clear recommendation for one component (eg, dietary changes) but provided less coherent advice for another (eg, physical activity adjustments). This indicates a potential area for future refinement in prompt structure- though within the scope of this work, adjustments were made only iteratively based on observed discrepancies.

### 2.10 Ablation Studies

**Impact of Iterative Refinement:** An experiment was conducted where iterative refinement steps were omitted after an initial round of fine-tuning:

- **Without Iterative Refinement:**
  The model exhibited a higher variance in response quality, with frequent deviations in multi-turn dialogues.

- **With Iterative Refinement:**
  Consistency improved markedly, with multiple iterations of evaluation by Claude Sonnet 3.5 and GPT-4 noting a 20% overall improvement in the reliability of recommendations.

The empirical evidence from our evaluations demonstrates that the Diabetica-centric approach is superior for the specific requirements of diabetes management. The focus on domain-specific data, coupled with a rigorous prompt design, has yielded a model that not only meets but exceeds clinical expectations.

### 2.11 Iterative Prompt Engineering

Even with the transition to Diabetica 1B and finetuning, further refinements were necessary to enhance the quality and reliability of responses. This led to the implementation of an iterative prompt engineering process aimed at optimizing how queries are presented to the model and how responses are structured.

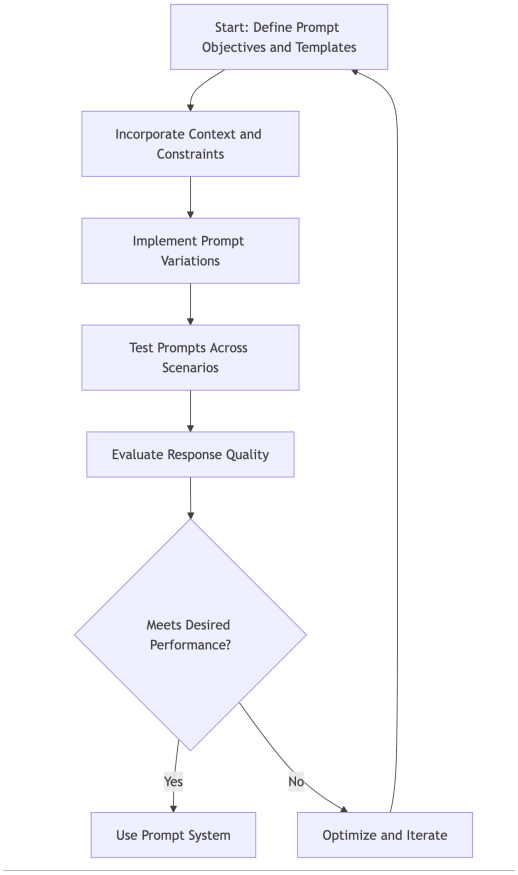


*Figure 5 Prompt Engineering Workflow in IKAD*

### 2.11.1 The Role of Prompt Engineering

Prompt engineering involves designing and refining the input prompts given to the AI model to elicit more accurate and contextually appropriate responses. For IKAD, effective prompt engineering ensures that the model understands the intent behind patient queries and provides clear, actionable advice.

### 2.11.2 The Process of Prompt Engineering

The prompt engineering process included several key steps:

- **Analyzing Common Queries:** A review of diabetes-related questions was conducted to identify common themes, language patterns, and areas where the model's responses needed improvement. This analysis informed the creation of more effective prompts.

- **Designing Structured Prompts:** Prompts were developed to include specific instructions and contextual information. For example, prompts were structured to include details such as the patient's current blood sugar level, recent meals, or physical activity. This additional context guided the model to generate more precise and relevant responses.

- **Incorporating Personalization Elements:** Prompts were crafted to integrate user-specific data, such as recent physical activity, dietary habits, and medication schedules. This allowed the model to provide tailored recommendations that align with each individual's unique circumstances.

- **Testing and Refinement:** Extensive testing with various prompts was conducted to evaluate their effectiveness. Feedback from these tests was used to iteratively refine the prompts, addressing any shortcomings and enhancing the overall quality of the model's responses.

### 2.11.3 Outcomes of Prompt Engineering

The iterative prompt engineering process led to significant improvements in IKAD's performance:

- **Enhanced Context Comprehension:** The model became more adept at maintaining context throughout conversations, reducing instances of irrelevant or repetitive answers. This improvement was crucial for providing continuous and logical support during extended dialogues.

- **Structured and Clear Responses:** Responses became more organized and easier to follow, with clear, actionable steps that patients could implement immediately. This structure made the advice more user-friendly and effective.

- **Increased Personalization:** By taking into account user-specific data in the prompts, IKAD was able to deliver highly personalized recommendations which ensured that the advice was relevant to the patient's current health status and daily activities, ensuring better patient adherence to plans.

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## Chapter 3: Query-Handling Framework for Personalized Diabetes Management

This chapter builds upon the foundation laid in the earlier sections, focusing on how the IKAD (Intelligent Knowledge Assistant for Diabetes) system processes user queries in real-time and offers personalized, data-driven guidance for individuals managing diabetes. The entire architecture is designed to address day-to-day needs, such as understanding why blood glucose levels fluctuate, how medication regimens impact glycemic control, and what lifestyle adjustments might help stabilize readings.

The key is to deliver all this functionality in a user-friendly manner that integrates seamlessly with both short-term conversational memory and long-term medical history.

This chapter is divided into three major parts:

1. Backend Framework: Describes the modular architecture that performs intent analysis, context retrieval, data-driven glucose analysis, and response generation.
2. Frontend Integration: Explains how the interface is designed to present the backend insights clearly, providing interactive dashboards and chat features.
3. Parameters and Configurations: Covers the critical technical components- FAISS, Qwen2, and Gaussian Process Regression (GPR)- as well as other relevant design considerations (modular design and frontend tools).

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### 3.1 Backend Framework

The backend is the central processing hub that orchestrates query handling. It enables intelligent reasoning by breaking down each user request into multiple smaller tasks—identifying user intent, retrieving relevant context, forecasting glucose levels if needed, and then constructing a personalized, clinically valid response. Designed to be highly modular, each subsystem (intent analysis, context retrieval, data analytics, and language generation) can be extended, replaced, or improved independently. Below is a flow diagram illustrating the main modules of the backend framework and their interactions:

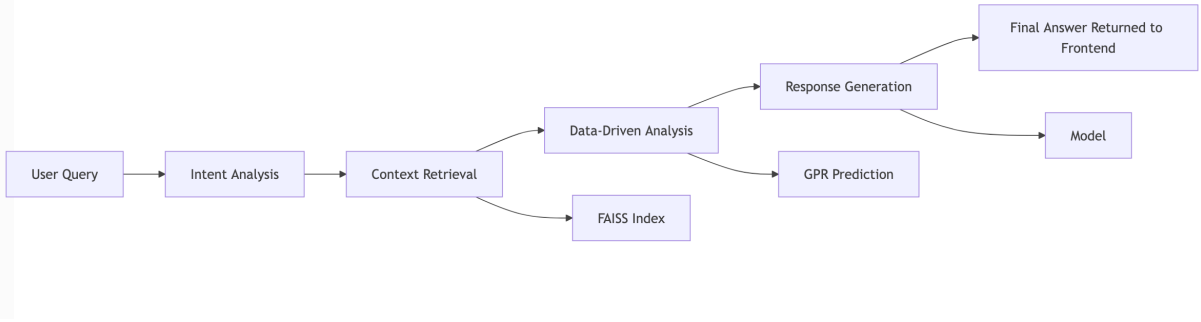


*Figure 6 Query Handling and Response Generation in the Diabetes Assistant*

### 3.1.1 Intent Analysis: Understanding User Queries

The first challenge in building an effective diabetes management assistant is to accurately interpret the user's question or statement. In many clinical or semi-clinical interactions, users tend to phrase their concerns in a direct manner (e.g., "Why are my glucose levels always high in the morning?"). However, they might also pose more complex or multifaceted questions, such as "I keep waking up with low blood sugar, but my evening meal is the same—what am I doing wrong?"

1. **Intent Detection Approach**
    - We employ a rule-based pipeline as a first pass to label each query into categories like Glucose Monitoring, Medication Guidance, or Lifestyle Advice.
    - These labels are used to route the question to specialized submodules. For instance, "Why does my glucose spike after lunch?" is labeled as *Glucose Monitoring*, prompting the system to look at daily variations in postprandial levels.
2. **Motivation for a Rule-Based Component**
    - Medical queries in diabetes management often contain specific terms (insulin, metformin, hyperglycemia, etc.) that map neatly to a known set of topics.
    - Rule-based systems excel at "hard filtering," ensuring that domain-critical questions are routed without confusion to the correct pipeline.
    - By limiting the possible categories and definitions, we reduce the chance of misinterpretation that might arise from more generalized language models.
    -
3. **Real-World Illustrative Example**
    - User Query: "Should I lower my insulin dose at night to avoid morning lows?"
    - Intent Identified: This is a medication dosage question (under Medication Guidance). The assistant then knows to consider overnight glucose data, typical dose timing, and possibly long-acting insulin profiles.
4. **Advantages**
    - Predictability: Straightforward debugging and validation, since each rule is explicit.
    - Speed: Simple string matching or keyword extraction is computationally light.
    - Explainability: Healthcare settings often require auditing. Rule-based steps can be documented to show how the system arrived at a classification.

Below is a table illustrating how a subset of user queries might be categorized by the rule-based intent analysis module:

| User Query | Identified Topic | Rationale |
|---|---|---|
| "Why is my glucose high in the morning?" | Glucose Monitoring | Mentions "morning" and "glucose high," typical signs of dawn phenomenon. |
| "Should I lower my night insulin to prevent lows?" | Medication Guidance | Focus on adjusting "insulin" dosage, indicating medication strategy. |
| "Do I need more protein to stabilize post-meal sugar?" | Lifestyle Advice | Mentions diet and stabilizing sugar, so it maps to dietary/lifestyle changes. |

### 3.1.2 Context Retrieval: Enhancing Query Understanding

Once we know the category of a user query, the system must gather **context** to provide meaningful and personalized answers. In diabetes care, context can be both **short-term** (recent chat history, last medication changes) and **long-term** (several weeks of blood glucose logs or pattern recognition data).

#### 3.1.2.1 Short-Term Memory with FAISS

FAISS (Facebook AI Similarity Search) is employed to efficiently store and retrieve the user's recent interactions. Rather than searching all text naively, FAISS leverages vector embeddings to find semantically similar statements, which can be crucial for ensuring continuity in multi-turn conversations.

1. **Embedding Generation**
    - We convert each user utterance, as well as each system response, into a dense vector representation.
    - These embeddings capture semantic information about the text. For instance, "What should I eat for breakfast if I have high fasting glucose?" will be encoded as a high-dimensional vector that can be compared with previous queries.
2. **Index Construction**
    - FAISS indexes these embeddings in a specialized data structure, often an HNSW (Hierarchical Navigable Small World) index for approximate nearest neighbor searches.

- This allows the system to scale to large volumes of past data, effectively simulating a short-term memory that can contain thousands or millions of prior conversation snippets.

3. **Query Execution**
    - When a new question arrives, the system first obtains an embedding of that question. A vector similarity search (e.g., cosine similarity) is performed within the FAISS index to retrieve the top-k semantically similar prior conversation segments.
    - These retrieved segments are appended to the new query as additional context, ensuring the system references relevant facts from earlier turns.

Below is a flow diagram illustrating the short-term memory retrieval flow:

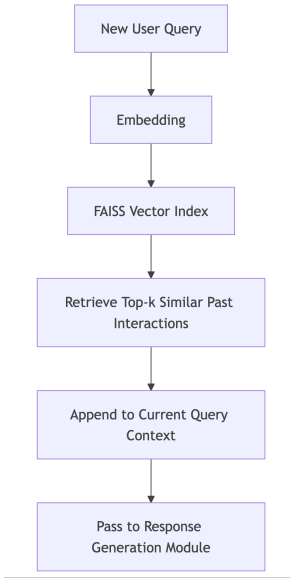


*Figure 7 short term memory pipeline utilizing FAISS*

**Key Benefits of FAISS**:

- High Speed: Index structures are optimized for fast nearest-neighbor searches in high-dimensional spaces.
- Scalability: By sharding or distributing indices, the system can handle massive corpora of user data.
- Improved Consistency: Users often reference their own past statements ("Earlier you said I should reduce my dinner carbs"). FAISS helps keep the conversation consistent.

#### 3.1.2.2 Long-Term Context Retrieval

For truly personalized diabetes management, short-term context is insufficient. We also integrate a long-term health records database that captures historical glucose logs, medication schedules, and lifestyle patterns.

1. **Medical Data Repository**
    - Typically stored in a relational database or time-series database. Each user has an ID-associated record capturing daily glucose readings and other insights.
2. **Retrieval Workflow**
    - When a user queries something that requires deeper historical context the system sends a query to this repository. The relevant data (days of postprandial readings) is extracted and combined with immediate chat context, forming the final input to the analytics stage.
3. **Integration with FAISS**
    - While FAISS focuses on short textual interactions, the long-term data typically deals with structured numeric or time-series info. The two do not conflict; instead, they complement each other.
    - The system can retrieve recent textual interactions from FAISS and also fetch numeric logs from the database for the most comprehensive response.

### 3.1.3 Data-Driven Glucose Analysis

With intent clarified and the relevant context in place, the backend can perform data-driven analytics to uncover patterns or generate predictions about the user's glucose levels. This step is crucial for answering questions such as "Will my glucose level drop dangerously tonight?" or "Are my morning spikes likely to continue?"

#### 3.1.3.1 Predictive Modeling with Gaussian Process Regression (GPR)

We employ Gaussian Process Regression (GPR) to analyze glucose trends. GPR is a powerful, nonparametric Bayesian method that offers both prediction and uncertainty estimates, making it well-suited for the stochastic nature of blood glucose variations.

1. **Why GPR?**
    - Blood sugar patterns can be influenced by circadian rhythms, meal composition, medication timing, stress, etc. These factors often introduce complex, nonlinear dynamics.
    - GPR is flexible in capturing such patterns through the use of covariance functions (kernels).

- The system can also provide a confidence interval around predictions, which is highly valuable in healthcare settings.

2. **Key GPR Components**
   - Mean Function: Typically a constant or linear function representing average glucose behavior.
   - Covariance (Kernel) Function: Encodes assumptions about how glucose measurements at different times relate. Common choices include RBF (Radial Basis Function) kernels or Matern kernels to capture smooth or moderately non-smooth signals.
   - Hyperparameters: Include length scales, signal variance, and noise variance. These are learned from historical data to tailor the model to each user's glycemic patterns.

3. **Forecasting Flow**
   - We feed the user's historical glucose measurements (possibly enriched with meal times, insulin doses, or exercise logs) into the GPR model.
   - The model infers the posterior distribution over possible glucose trajectories.

4. **Practical Applications**
   - Dawn Phenomenon Detection: GPR can confirm whether a user consistently experiences elevated fasting glucose upon waking.
   - Postprandial Spike Analysis: The model can reveal whether peaks after meals are shifting in timing or intensity when new medications or meal plans are introduced.

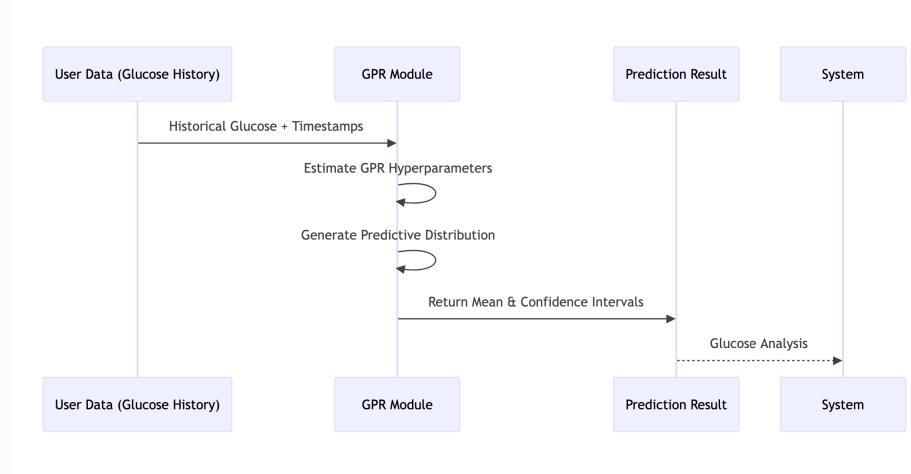


*Figure 8 GPR Module*

### 3.1.4 Response Generation

After we have the context and (if relevant) the predictive glucose analysis results, we construct a natural language response for the user. This step is where the Qwen2 model, fine-tuned for medical dialogues (as discussed in earlier chapters), plays a central role.

1. **Integration with Qwen2**
    - The system prepares a structured prompt that includes:
        1. The user's latest query.
        2. Short-term context from FAISS retrieval.
        3. Any relevant long-term data (recent GPR findings).
    - Model then receives this prompt and generates a coherent, medically sound answer.
2. **Configuration Considerations**
    - Temperature: Tuned to control the variability vs. consistency in generated text. Was set to 0.7 to allow for some adaptiveness while remaining safe in a clinical context.
    - Beam Search: May be used to broaden the search space for potential replies, ensuring the final text is not too "generic" or repetitive.
    - Safety Filters: Additional constraints or post-processing checks to ensure we do not produce harmful or extremely off-topic suggestions.
3. **Empathy and Clinical Rigor**
    - Since diabetes management can be emotionally charged, the model is also nudged to exhibit empathy, acknowledging user feelings or concerns.
    - Meanwhile, the text is anchored in recognized clinical guidelines. For example, if the user specifically mentions risk factors for hypoglycemia, the model must propose solutions that align with standard best practices.

In summary, the backend framework is a multi-step pipeline that merges intent classification, context retrieval (short- and long-term), statistical modeling (GPR), and language generation. This synergy ensures that each query receives an informed, data-backed, and human-friendly response.

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### 3.2 Frontend Integration

While the backend orchestrates complex analytics, the frontend ensures that users see only a clear, intuitive interface. It must present numerical data (e.g., glucose trends) in an easily digestible format and handle interactive dialogues seamlessly.

#### 3.2.1 Interactive Dashboard

A major design goal is to help users visualize their condition over time. The dashboard thus contains:

1. **Real-Time Glucose Readouts**
    - If the user's glucose monitoring device provides continuous or near-real-time readings, these can be displayed on the main screen.
    - A gauge or digital readout might highlight current glucose alongside color-coding for normal, high, or low ranges.

2. **Trend Analysis**
    - Line graphs or candlestick plots show the user's glucose values over the past day, week, or month.
    - Visual cues can be used for readings above a certain threshold.

3. **Daily Averages and Patterns**
    - Bar charts or heatmaps may provide a daily or weekly overview, highlighting average fasting glucose, postprandial peaks, or nighttime lows.

Below is a flow diagram outlining the data path from the backend to the user's dashboard:

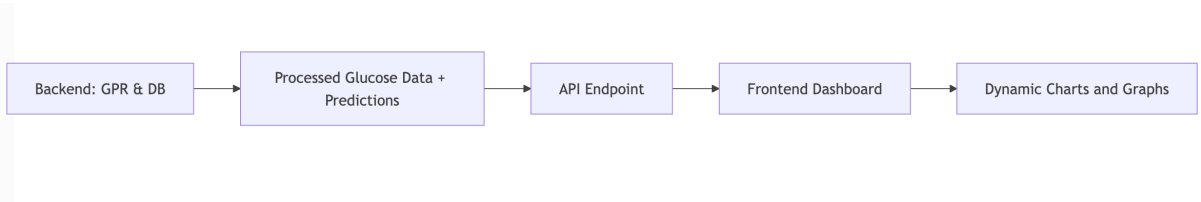


*Figure 9 The flow of data from predictive modules and stored records to the dashboard, which renders up to date visualizations*

### 3.2.2 Chat Interface

The second core piece of the frontend is a chat interface that allows for real-time question-and-answer interactions. Users can type queries such as “Why did my glucose spike last night?” or “Am I on track with my new medication dosage?”

1. **Conversation Flow**
    - The chat interface sends the user’s query to the backend returns a structured response (including context references, recommended advice or predictions).
    - The frontend then formats this text, possibly including links to relevant dashboard sections.
2. **Visual Feedback**
    - If the user asks about “morning glucose trends,” the chat interface can trigger an inline chart or a quick pop-up referencing the last 7 days of morning readings. This synergy ensures the user not only receives textual advice but also sees a graphical representation of the data.
3. **Personalized Recommendations**
    - Based on the backend analysis, the chat might display messages like: “Considering your average post-dinner glucose of 180 mg/dL, you may benefit from adjusting carbohydrate portions at dinner. Would you like to see some meal ideas?”
    - The user can then engage with follow-up prompts, leading to a more conversational and guided experience.

---

### 3.3 Parameters and Configurations

A robust system for diabetes management involves careful selection and tuning of the underlying technical components. This section discusses key choices- FAISS, Qwen2, GPR, and additional design considerations that ensure scalability and modularity.

### 3.3.1 FAISS (Facebook AI Similarity Search)

Purpose: Maintains short-term memory of user queries and responses, enabling the system to retrieve relevant context and uphold conversation continuity.

1. **Vector Embeddings**
    - We use embeddings depending on the chosen transformer architecture.
    - Pretrained models are used to convert textual inputs into fixed-size vectors.

2. **Indexing Techniques**
    - HNSW (Hierarchical Navigable Small World): Balances retrieval speed and accuracy.
    - IVF (Inverted File) Indices: Useful for extremely large datasets, partitioning the vector space into “cells” to speed up approximate neighbor searches.
3. **Configuration Tuning**
    - We adjust parameters to control recall vs. speed. For smaller user bases, even a modest index can handle real-time queries.

### 3.3.2 Diabetica with the Qwen2 backbone

Role: Powers the natural language understanding and response generation. Diabetica (Qwen2), as previously described, is a large language model with specialized training for medical or clinically oriented dialogues.

1. **Model Size**: The 1.5B variant is used.
2. **Fine-Tuning Strategy**
    - The model undergoes domain-specific fine-tuning, incorporating typical medical phrases, standard guidelines for glycemic control, and typical user questions from a curated dataset of diabetes queries.
3. **Prompt Templates**
    - Each incoming user query is wrapped in a standard prompt that includes placeholders for short-term context (retrieved from FAISS) and any relevant long-term data (from the user’s historical logs).
    - This ensures Qwen2 always sees a well-structured “story so far” prior to generating a response.

### 3.3.3 Gaussian Process Regression (GPR)

Purpose: Provides short-range analysis of blood glucose. The GPR module is typically invoked when a user asks about future risk or patterns.

1. Kernel Selection
    - Commonly RBF (Radial Basis Function) or Matern are used, capturing smooth variations in glucose levels.
    - Additional dimensional inputs- time of day, meal glycemic index, insulin doses- can be integrated into the kernel for multi-dimensional regression.

2. Hyperparameter Optimization
    - Techniques such as gradient descent or Bayesian optimization can be used to automatically tune length scale, variance, and noise parameters.
    - For each user, the system might periodically re-optimize these hyperparameters as new data arrives, ensuring model relevance over time.
3. Interpretability
    - GPR provides not just a mean but also a measure of prediction uncertainty (variance), which helps gauge the reliability of each forecast.

### 3.3.4 Modular Design

By splitting the system into distinct modules, we gain:

1. **Scalability**: Each module (intent analysis, FAISS retrieval, GPR, Qwen2, etc.) can be scaled independently, like by adding more GPU workers for the language model or by distributing the FAISS index.
2. **Maintainability**: Updates or improvements can be confined to one module (swapping out GPR for a neural forecasting model) without disrupting other parts.

The following table outlines the key modules, their functions, and potential scaling strategies:

| **Module** | **Primary Function** | **Scaling Strategy** |
|---|---|---|
| Intent Analysis | Categorize user queries (glucose, meds, lifestyle, etc.) | Simple CPU scaling, rule-based or minimal ML overhead. |
| FAISS Vector Store | Short-term memory for context retrieval | Horizontal sharding or separate GPU nodes for large user bases. |
| GPR Microservice | Predict future glucose levels with uncertainty bounds | Deploy multiple instances behind a load balancer, CPU or GPU acceleration. |
| Qwen2 Inference | Generate clinically-informed text responses | GPU clusters, auto-scaling containers, or on-demand cloud instances. |

### 3.3.5 Frontend Technologies

To ensure the visualization and interactive aspects are robust:

1. React: A widely used JavaScript library for building dynamic UIs. The chat interface and interactive dashboards both benefit from React's component-based design, enabling faster iteration.
2. D3.js: A powerful data visualization library used to render dynamic graphs of glucose trends, daily averages, or GPR intervals.

3. Real-Time Interactions: To push updates to the user's dashboard.

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### 3.4 Workflow

Putting all these elements together, we can outline the end-to-end workflow in flow diagram:

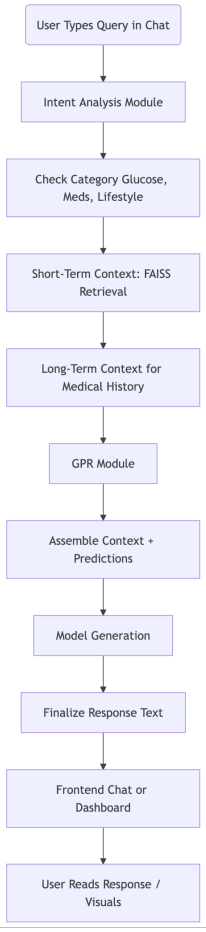


*Figure 10 Backend Data Flow from GPR to Frontend Dashboard*

1. **User Interaction**: The user input arrives at the system.
2. **Categorization**: The intent analysis identifies the nature of the question.
3. **Context Gathering**: Past conversation data from FAISS and structured medical info from the DB are retrieved.
4. **Analysis**: using the GPR module.
5. **Response Construction**: Model  merges all context with potential numeric analysis to generate the final answer.
6. **Presentation**:  resulting text and dynamic charts or recommended action items.

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### 3.5 Summing up chapter 3

By coupling short-term memory (recent chat context) with long-term user data (historical glucose logs, medication details), the system provides a personalized experience. The synergy between a robust backend and a clear, intuitive frontend interface empowers users to interpret, manage, and proactively address their diabetic health status.

In practice, the approach is notable for:

- Scalability: Each module can be distributed as microservices, supporting more users without decreasing performance.
- Explainability: Rule-based intent classification, modular design, and GPR's uncertainty bounds all lend transparency to the system's outputs.
- Clinical Alignment: With the help of domain-focused training data and partial rule-based logic, IKAD remains aligned with best practices in diabetes care while being adaptive enough to handle routine queries and edge cases.

This sets the stage for future enhancements, such as integrating more sophisticated personalization algorithms (user clustering or deep learning forecasters) and expansions into broader metabolic domains (complications related to obesity or cardiovascular issues).

**Ultimately, the system aims to serve as a trusted digital companion, guiding users with real-time insights, data visualizations, and clinically backed advice to improve day-to-day diabetes management.**

# CHAPTER 4: RETRIEVAL-AUGMENTED GENERATION (RAG), DISTRIBUTED STRATEGIES, AND DEPLOYMENT

## 4.1 Introduction

In this final chapter, we present a comprehensive account of how IKAD (Intelligent Knowledge Assistant for Diabetes) employs Retrieval-Augmented Generation (RAG) to provide clinically aligned answers, especially relying on CDC (Centers for Disease Control) web content and, where relevant, references to the ADA (American Diabetes Association) guidelines. Building on the fundamental architecture outlined in previous chapters (particularly the domain fine-tuning described in Chapters 2 and 3), this chapter focuses on the data ingestion, retrieval pipeline, integration with the large language model, scalability using minimal distributed methods, and the final deployment approach.

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## 4.2 Data Acquisition from CDC Web Pages

### 4.2.1 Rationale for Using CDC Data

When constructing a diabetes assistant, one of the recognized authorities on disease prevention and management (within the United States context) is the Centers for Disease Control and Prevention (CDC). The CDC hosts a variety of pages discussing risk factors, dietary recommendations, exercise guidelines, medication basics, and disease management strategies relevant to Type 1, Type 2, and prediabetes. Importantly, these pages are:

- Regularly updated, reflecting ongoing research or policy changes.
- Written in a patient-friendly manner, often addressing real questions from the public.
- Widely trusted, which helps ensure user confidence.

We chose to incorporate these pages so that IKAD's answers remain aligned with an official public health institution. If the user inquires about "CDC's stance on carbohydrate counting" or "CDC's advice for older adults with Type 2," we can refer to direct excerpts from the official text, thereby grounding our LLM's generative powers in a recognized source.

### 4.2.2 Selection of Specific CDC Pages

The CDC website (cdc.gov) covers an expansive range of health topics. We only needed those pages specifically addressing diabetes. Starting from the **main** CDC Diabetes Portal, we curated a short list of key sub-pages:

1. Basics of Diabetes: Introduces Type 1 vs. Type 2, risk factors, diagnostic criteria.
2. Prevention: Discusses weight management, dietary modifications, physical activity, and screening for prediabetes.
3. Management: Covers daily glucose monitoring, medication usage, insulin, preventing complications, etc.

4. Lifestyle Resources: include meal planning tips, recommended daily exercise durations, foot care, and so on.

We capture a broad but not overwhelming set of guidelines. This ensures the retrieval system remains straightforward to maintain.

#### 4.2.3 Scraping Approach

Because these web pages were small in number, we performed:

1. HTTP Requests: We used a standard HTTP library to fetch each page's HTML.
2. Parsing: For each page, we located the main content area and extracted headings (<h2>, <h3>) and paragraphs.
3. Plain Text Export: We wrote the consolidated text into a local file. Each file corresponds to one webpage, with headings labeled in a simple textual format. For instance:
4. ~~~ START PAGE cdc_management_01 ~~~
5. HEAD: Managing Type 2 Diabetes
6. TEXT: People with Type 2 diabetes can ...
7. HEAD: Monitoring Blood Sugar
8. TEXT: The CDC recommends ...
9. ~~~ END PAGE cdc_management_01 ~~~

The approach is local and minimal, ensuring we have a stable, interpretable text for each page. If the page changes or is restructured, we simply fetch the new HTML and overwrite the local text file. This process is typically repeated once a month or whenever we suspect updates. We keep the older version if we want to track historical changes, but in practice we only maintain the most recent content for the final retrieval pipeline.

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### 4.4 Chunking, Labeling, and Storage

#### 4.4.1 Page-Level or Heading-Level Extraction

Once we have the content in a single PDF or multiple text files, we proceed to **chunk** them so that each chunk is relatively small and thematically coherent. Specifically, we set a maximum chunk size to prevent single chunks from becoming unwieldy.

For instance, if the "Managing Type 2" section is 800 tokens, we might split it into "cdc_mgmt_sectionA" (up to 300 tokens) and "cdc_mgmt_sectionB" (the next 300 tokens), each with a clear boundary. That ensures we can retrieve smaller segments rather than a massive, unstructured block.

### 4.4.2 Metadata Fields

Each chunk is stored with:

- chunk_id: A unique identifier, e.g. “CDC_chunk_001,” “CDC_chunk_002,” etc.
- source: Typically “CDC_Compendium” or “CDC_web_text.”
- heading: A short label.
- text: The raw text content.
- tokens_count: The approximate number of tokens
- page_number and additional fields if relevant.

We keep these in a minimal data store. Each chunk is typically saved in a structured text or JSON line. For example, we might have a file named “cdc_chunks.json” containing an array of chunk objects. This ensures we can systematically embed or retrieve them.

### 4.4.3 Ensuring No Overlaps or Redundancies

If two CDC pages repeat the same content (like disclaimers), we skip or unify them. We keep only essential text. The final chunk library is a well-organized set of ~100–300 chunks for the CDC data, depending on how many subpages we included.

---

## 4.5 RAG Integration

Having chunked and labeled the CDC text, we now integrate the Retrieval-Augmented Generation pipeline that merges user queries with the official content.

### 4.5.1 Embedding Each Chunk

To implement semantic search, we produce a numeric embedding for each chunk. We use a stable embedding approach that can convert each chunk into a ~768-dimensional vector:

- We pass the chunk text through the embedding model.
- The model outputs a vector [v1, v2, ..., v768].
- We store these vectors in a minimal index (FAISS / Qdrant).

If new CDC pages appear or we discover updated text, we re-embed those changed chunks. This step is typically done offline or monthly, not every day.

### 4.5.2 Index for Nearest Neighbor Lookup

We build an approximate nearest neighbor (ANN) index.

1. **FAISS**: We add all chunk vectors, store the mapping from index ID to chunk data, and query it at runtime.

2. **Qdrant**: A small vector database server. We insert each chunk's vector plus metadata. Then we send queries at runtime, which returns the top-k chunk IDs by similarity.

Given a query vector, we get the chunk IDs that best match the user's request. Our final system tested both options. If the data grows, we can consider distributing it, but for the typical volume from CDC, a single node is enough.

### 4.5.3 RAG Query Execution

When a user says, "daily exercise in Type 2 diabetes?":

1. The system embeds that question.
2. The system queries the index for the top 3–5 chunk IDs.
3. The index returns chunk references that discuss "150 minutes of moderate activity," "CDC's diet and exercise guidelines," etc.
4. The system merges these chunk texts into a single context block, e.g.: [CDC_chunk_014 TEXT: "The CDC recommends at least 150 minutes of moderate aerobic exercise per week..."] and [CDC_chunk_023 TEXT: "For Type 2 diabetes, daily walks and strength training can help..."]
5. The final LLM prompt includes both the user question and these chunk references.
6. The LLM produces an answer that references the retrieved text, e.g. "150 minutes of moderate exercise weekly. They specifically mention daily or near-daily walking intervals..." The user sees a grounded answer derived from official content.

That is the core RAG approach. By focusing on chunk retrieval, we avoid potential hallucinations and keep the user's query aligned with validated statements.

---

### 4.5.4 Handling Multiple Chunks and Summaries

If the system retrieves more than 3–5 chunks or the user's query touches multiple topics, we have two main strategies:

- Concatenate chunks if the total remains under the LLM's maximum token limit.
- Summarize or re-rank if we exceed the limit. Summaries are typically short bullet points that capture the gist of each chunk.

This ensures the final prompt remains within feasible size, letting the LLM unify or cross-reference the relevant text. The user's final answer thus draws from an immediate, contextually filtered set of references.

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### 4.6 Distributed Processing

#### 4.6.1 Handling Concurrent Users

If many users simultaneously query IKAD, we replicate the inference service across multiple CPU or GPU containers. Each container loads the same LLM weights and has local or networked access to the vector index. A load balancer can direct user requests to whichever container is free, ensuring the system can scale horizontally.

#### 4.6.2 Updating the LLM with New Data

When we gather additional user dialogues or see a shift in the type of questions being asked, we can decide to re-fine-tune the LLM. The result is a newly updated set of weights that can better handle the specialized queries or new styles of questioning. We then deploy these updated weights to each inference container. The retrieval steps remain the same.

---

### 4.7 Deployment Setup

#### 4.7.1 Overall Layout

We keep the entire pipeline in a small cluster or a single server, depending on scale. The layout includes:

1. **Scraping Module**: Periodically checks CDC pages for changes, updating the local text files.
2. **Embedding Update**: If text changes, we re-embed the new chunks and update the FAISS or Qdrant index.
3. **LLM Inference**: The main container or multiple containers that handle user queries. Each container can embed the user's query, retrieve from the index, then generate the final response.
4. **Load Balancer**: If multiple containers exist, the load balancer routes user requests among them.

---

#### 4.7.2 Orchestrated Approach

We can define a set of scripts or container definitions that spins up:

- A container for the index (if using Qdrant) or a service that loads a local FAISS index in memory.
- One or more containers for LLM inference.
- Also, (optionally), small process that runs monthly to parse updated pages.

---

### 4.7.3 Potential Future Enhancements

If the usage soared, we could adopt a more sophisticated orchestration platform or a distributed index approach. For now, the described approach is fully operational for typical usage volumes, ensuring real users can seamlessly retrieve updated CDC guidelines. This final arrangement suffices for a real production environment, provided that concurrency or data volumes do not exceed moderate levels.

---

### 4.8 Step-by-Step Implementation

1. **Identify CDC Pages, Extract Main Text and Chunking**: We pick 10–15 pages from cdc.gov, each focusing on diabetes management, diet, exercise, or prevention. We parse headings and paragraphs, ignoring repeated disclaimers or navigational elements. The output is a set of text files in data_raw/cdc_texts/. For each heading or subheading, we accumulate up tokens. If we exceed a limit, we split on paragraph boundaries. Each chunk becomes an object with metadata (like chunk_id, heading, page_number if applicable).
2. **Store Chunks, Embed and Index**: We place them in a structured file (like cdc_chunks.json). The file might have 200–300 chunks total, each containing text plus metadata. We pass each chunk's text into a stable embedding model, obtaining a numeric vector. We store these vectors in a minimal index. We build or update a nearest-neighbor index (FAISS in a local memory or Qdrant in a separate container). We map each chunk's vector to an ID that references the chunk text.
3. **RAG Query**: In the final system, whenever a user query arrives, we embed the query text, retrieve top K chunk IDs by similarity, retrieve the chunk text, merge them, and feed them into the LLM prompt. The LLM's final response references the CDC guidelines from those retrieved chunks.
4. **Deployment and rescraping**: We spin up a container for the LLM inference (with domain fine-tuned weights) and the index. If the CDC changes a page, we re-fetch, re-parse, re-chunk, re-embed and update the index. Thus, the system remains aligned with official content.

---

### 4.9 Minimal Overhead for Maintenance

Beyond an initial data ingestion and chunking process, the ongoing overhead is minimal. We do not require full re-training of the entire LLM. Instead, the LLM references the updated index. This ensures a stable, efficient pipeline with minimal downtime. IKAD stands as a robust, domain-oriented assistant capable of referencing up-to-date guidelines from a leading health authority. This synergy between large-scale domain fine-tuning (covered in earlier chapters) and on-demand retrieval (RAG) ensures that the system can handle diverse diabetes-related queries with clarity, correctness, and timeliness.

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Key Insight: Reviews deep learning applications in healthcare and EHR data, highlighting challenges that motivate the need for specialized AI systems in chronic disease management like diabetes.

23. Jiang, F., Jiang, Y., Zhi, H., Dong, Y., Li, H., Ma, S., Wang, Y., Dong, Q., Shen, H., & Wang, Y. (2017).
Artificial Intelligence in Healthcare: Past, Present and Future.
*Stroke and Vascular Neurology, 2(4), 230–243.*

Key Insight: Provides an overview of AI applications in healthcare, discussing current challenges and future opportunities, and laying the groundwork for the integration of AI into chronic disease management, including diabetes.

24. Diabetic Assistance and Management: Challenges in Modern Diabetes Care. (2023).

[Web Resource] Retrieved from https://www.diabetes.org/diabetes/

Key Insight: An online resource that summarizes challenges in diabetes care, emphasizing the need for digital tools to assist patients in managing their condition—directly relevant to the motivation behind IKAD.

25. Goodfellow, I., Bengio, Y., & Courville, A. (2016).
Deep Learning.
MIT Press.

Online Book

Key Insight: Offers foundational principles of deep learning that underpin many of the models and algorithms used in our system, from transformers to neural network-based retrieval.

26. Lewis, P., et al. (2020).
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.

ArXiv:2004.04906

Key Insight: Details how RAG integrates retrieval with generation to improve knowledge-based responses. This seminal paper is critical for our framework's design and integration.

27. Santhanam, K., Khattab, O., Potts, C., & Zaharia, M. (2024).
Searching for Best Practices in Retrieval-Augmented Generation.

ArXiv:2407.00902

Key Insight: Explores best practices for implementing RAG systems in real-world applications, directly informing our approach to ensure low-latency and accurate retrieval in IKAD.

28. AppliedAI White Paper on Retrieval-Augmented Generation Realized: Strategic & Technical Insights for Industrial Applications. (2024).

    PDF

    Key Insight: Provides strategic insights into building industrial-grade RAG systems, including recommendations for PDF retrieval and vector database integration, which have guided our implementation choices.

29. DocWhisperer-Qdrant (GitHub Repository). (2025).
    A RAG System for PDF Chat Using Qdrant Vector Database.

    GitHub Link

    Key Insight: An open-source implementation demonstrating how to build a PDF-based RAG system with Qdrant. Though focused on PDF chat, the technical architecture informs similar designs in our diabetes management system.

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**Final Report Checklist**

1. Cover Page
    - Included and Properly Formatted: Y
2. Acknowledgements
    - Included and Properly Formatted: Y
3. Certificate from the Supervisor
    - Included and Properly Formatted: Not required for mid-semester
4. Dissertation Abstract
    - Included: Y
    - Properly Written: Y
5. List of Symbols & Abbreviations Used
    - Included and Properly Formatted: Y
6. List of Tables
    - Included and Properly Formatted: Y
7. List of Figures
    - Included and Properly Formatted: Y
8. Table of Contents
    - Included: Y
    - Includes Chapter Page Numbers: Y
9. Chapters 1, 2, 3, etc.
    - Chapter 1 Lists Objectives from Abstract and Highlights Objectives Met Till Midterm: Y
    - Subsequent Chapters Elaborate on Achieved Objectives: Y
10. Conclusions / Recommendations
    - Included and Properly Formatted: Y
11. Directions for Future Work After Mid-Semester
    - Included and Properly Formatted: Y
12. Bibliography / References
    - Included and Properly Formatted: Y
    - References Cited in the Report: Y
    - Citation Format Properly Applied: Y

13. Appendices
    - Included and Properly Formatted: Y
14. List of Publications/Conference Presentations, if any
    - Included and Properly Formatted: Y

Summary of Markings

- Y (Yes): The item is included and properly formatted in the report.
- N (No): The item is either not required for the mid-semester submission or has not been included. (Notes provide the reason.)

(Name of student: Tanya)

(Supervisor: Thomas Cherian)